UNIVERSITY OF MUMBAI

ST. GONSALO GARCIA COLLEGE OF ARTS AND COMMERCE

Vasai (West), Dist. Palghar – 401 201, Maharashtra



DEPARTMENT OF INFORMATION TECHNOLOGY

A COMMUNITY ENGAGEMENT PROJECT (CEP) REPORT ON

WASTEIQ: AI-POWERED WASTE CLASSIFICATION AND COMMUNITY RECYCLING AWARENESS SYSTEM

Submitted in partial fulfillment of the requirements for the Degree of

BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)

(As per National Education Policy - NEP 2020 Guidelines with effect from AY 2026-2027)

SUBMITTED BY:

Mr. Yadav Krishna Panchdev

Roll No: **98**

(Project Partner: Mr. Patel Deepak Rajendra, Roll No: 68)

Class: T.Y. B.Sc. (Information Technology)

Semester: Semester V

UNDER THE GUIDANCE OF:

Prof. Olivia Dias

Project Mentor

Department of Information Technology

St. Gonsalo Garcia College, Vasai (W)

MONTH OF SUBMISSION: NOVEMBER 2026 | ACADEMIC YEAR: 2026 – 2027

DEPARTMENT OF INFORMATION TECHNOLOGY

ST. GONSALO GARCIA COLLEGE OF ARTS AND COMMERCE

Vasai (West), Dist. Palghar – 401 201, Maharashtra

DEPARTMENT OF INFORMATION TECHNOLOGY

CERTIFICATE OF APPROVAL

This is to certify that the Community Engagement Project (CEP) titled **"WASTEIQ: AI-POWERED WASTE CLASSIFICATION AND COMMUNITY RECYCLING AWARENESS SYSTEM"** is a bonafide work carried out and submitted by **Mr. Yadav Krishna Panchdev**, Roll No: **98**, in collaboration with project partner **Mr. Patel Deepak Rajendra**, Roll No: **68**, students of **T.Y. B.Sc. (Information Technology)**, **Semester V**, during the Academic Year **2026–2027**.

This report is submitted in partial fulfillment of the requirements for the award of the Degree of **Bachelor of Science in Information Technology** under the **University of Mumbai**, as prescribed under the National Education Policy (NEP 2020) curriculum guidelines (Circular UG/9.1/2025-26).

To the best of our knowledge, the work carried out by the student is original, and the information, user demonstration records, and software implementation details included in this project report are authentic, correct, and collected as per the NEP 2020 guidelines prescribed by the University of Mumbai.

Prof. Olivia Dias Project Mentor	**Head of Department** Department of Information Technology
External Examiner University of Mumbai	**Dr. Somnath Vibhute** Principal / Director

College Seal / Stamp:

Academic Year: 2026–2027

ST. GONSALO GARCIA COLLEGE OF ARTS AND COMMERCE
DEPARTMENT OF INFORMATION TECHNOLOGY

CERTIFICATE BY THE PROJECT MENTOR

I hereby certify that **Mr. Yadav Krishna Panchdev** (Roll No: **98**), in active collaboration with project partner **Mr. Patel Deepak Rajendra** (Roll No: **68**), student of **T.Y. B.Sc. (Information Technology)**, **Semester V**, has carried out the Community Engagement Project titled **"WASTEIQ: AI-POWERED WASTE CLASSIFICATION AND COMMUNITY RECYCLING AWARENESS SYSTEM"** under my direct supervision, guidance, and mentorship during the Academic Year **2026–2027**.

The candidate has successfully developed the WasteIQ web application and conducted interactive demonstrations and usability evaluations with college classmates and neighborhood peers. The student demonstrated active engagement in applying machine learning and web technology toward community solid waste awareness.

The report submitted is an authentic piece of software implementation and peer-testing synthesis. I recommend this report for evaluation toward the two-credit requirement of the Community Engagement Project under NEP 2020.

Prof. Olivia Dias
Project Mentor
Department of Information Technology
St. Gonsalo Garcia College of Arts and Commerce, Vasai (West)
Dist. Palghar – 401 201

Place: Vasai (West), Maharashtra | **Academic Year:** 2026–2027

STUDENT'S DECLARATION

(In accordance with Annexure IV of University of Mumbai CEP Guidelines UG/9.1/2025-26)

I, **Mr. Yadav Krishna Panchdev** (Roll No: **98**), student of **T.Y. B.Sc. (Information Technology)**, **Semester V**, studying at St. Gonsalo Garcia College of Arts and Commerce, Vasai (West), hereby declare that the Community Engagement Project (CEP) report entitled:

> **"WASTEIQ: AI-POWERED WASTE CLASSIFICATION AND COMMUNITY RECYCLING AWARENESS SYSTEM"**

is an original, bonafide, and independent piece of work carried out by me in collaboration with my project partner **Mr. Patel Deepak Rajendra** (Roll No: **68**), under the supervision and guidance of **Prof. Olivia Dias**, Department of Information Technology, St. Gonsalo Garcia College, during the Academic Year **2026–2027**.

I solemnly affirm and state that:

1. The system implementation details, live mobile testing logs, and peer user feedback presented in this report are genuine records of work conducted with our college classmates and neighborhood friends in Vasai.
2. The web platform and computer vision model (*WasteIQ*) described in this document was independently architected, developed, and evaluated for public community benefit.
3. The matter embodied in this report has not been submitted previously to this or any other university or institution for the award of any degree, diploma, or certificate.
4. Standard ethical practices have been observed, ensuring the privacy and consent of all participating peers. All scholarly references are acknowledged in standard APA format.

Mr. Yadav Krishna Panchdev
Roll No: 98
Class: T.Y. B.Sc. (Information Technology)
Semester: Semester V
St. Gonsalo Garcia College, Vasai (West)

Place: Vasai (West), Maharashtra
Academic Year: 2026–2027

ACKNOWLEDGEMENT

The completion of this Community Engagement Project (CEP) titled **"WasteIQ: AI-Powered Waste Classification and Community Recycling Awareness System"** represents a deeply fulfilling journey of combining technical software engineering with practical environmental awareness. I take this opportunity to express my heartfelt gratitude to all individuals and institutions who extended their guidance and encouragement throughout this project during the Academic Year 2026–2027.

First and foremost, I express my profound gratitude and respect to my project mentor, **Prof. Olivia Dias**, Department of Information Technology. Her insightful technical guidance, continuous encouragement, constructive reviews, and guidance during system design and testing made this project possible.

I extend my sincere thanks to the Department of Information Technology and faculty members for their administrative support, encouragement, and for providing the departmental computing resources required for developing the web platform.

I am deeply grateful to **Dr. Somnath Vibhute**, Principal of St. Gonsalo Garcia College of Arts and Commerce, Vasai (West), for fostering an academic environment that promotes societal service, ecological awareness, and technological innovation as mandated under the National Education Policy (NEP 2020) framework.

I would like to extend my heartfelt appreciation to my project partner, **Mr. Patel Deepak Rajendra** (Roll No: 68), for his exceptional technical collaboration, dedication, and expertise in developing the Google Gemini multimodal vision pipeline, real-time camera interfaces, AI chatbot assistant, and responsive frontend architecture throughout the project lifecycle.

I also express my sincere appreciation to our college classmates and neighborhood friends in Vasai who enthusiastically participated in testing the WasteIQ web application on their mobile devices and provided constructive feedback.

Place: Vasai (West), Maharashtra

Mr. Yadav Krishna Panchdev
Roll No: **98**
T.Y. B.Sc. (Information Technology) — Semester V

ABSTRACT

Solid waste mismanagement and lack of segregation at source pose severe environmental and public health hazards in modern metropolitan and suburban regions. In rapidly growing suburban centers such as Vasai-Virar, rapid urban expansion has led to elevated volumes of municipal solid waste, where food refuse, recyclable dry plastics, electronic components, and domestic hazardous items are frequently dumped into mixed garbage bins. This absence of segregation leads to overflowing dumping sites, toxic groundwater leachate, and substantial loss of valuable recyclable materials.

To address this issue through accessible software, this Community Engagement Project (CEP)—undertaken under the National Education Policy (NEP 2020) curriculum prescribed by the University of Mumbai—presents **WasteIQ**. WasteIQ is an intelligent, responsive web platform engineered to assist citizens, households, and students in classifying household waste items instantly and fostering responsible recycling habits through real-time camera scanning, resin code guidance, and localized e-waste awareness.

The software leverages the Google Gemini Multimodal Vision API integrated with Next.js 14, Tailwind CSS, and Clerk Authentication. Users capture or upload an image of any waste article using standard smartphone cameras. In under two seconds, the AI vision pipeline categorizes the item into one of four standard municipal bins: **Wet/Organic (Green)**, **Dry/Recyclable (Blue)**, **E-Waste (Purple)**, or **Domestic Hazardous (Red)**. The system simultaneously displays pre-disposal preparation instructions, plastic resin codes (SPI 1–7), and eco-points.

The web platform was evaluated through hands-on testing sessions conducted with a group of 11 participants, comprising college classmates from the Department of Information Technology and local neighborhood friends in Vasai. Participants accessed the deployed application on their personal mobile devices, testing items such as plastic bottles, milk packets, kitchen food scraps, chargers, and batteries. User feedback validated fast classification response, clear color-coded guidance, and high utility in resolving common household disposal confusion.

Keywords: WasteIQ, Waste Classification, Google Gemini Vision AI, Source Segregation, Community Engagement Project (CEP), NEP 2020, E-Waste Management, User Testing, St. Gonsalo Garcia College, Vasai.

TABLE OF CONTENTS

Chapter / Section	Title	Page No.
—	Title Page	1
—	Certificate by the Institute (Certificate of Approval)	2
—	Certificate by Mentor	3
—	Student’s Declaration	4
—	Acknowledgement	5
—	Abstract & Keywords (200–300 Words)	6
—	Table of Contents	7
—	List of Figures and Tables	8
CHAPTER 1	**INTRODUCTION**	**9**
1.1	Purpose of the Project & Core Objectives	9
1.2	Background Information of Waste Mismanagement in Vasai Region	9
1.3	Scope of the Report & Technical Boundaries	10
1.4	Relevance to NEP 2020 & UN Sustainable Development Goals	10
CHAPTER 2	**LITERATURE REVIEW**	**11**
2.1	Review of Regional SWM Regulations & Municipal Rules	11
2.2	Role of AI and Mobile Web Technology in Recycling Awareness	11
2.3	Community Engagement Models & Gamification Pedagogy	12
2.4	Plastic Resin Identification Standards (SPI Codes 1 to 7)	12
CHAPTER 3	**METHODOLOGY & INDIVIDUAL CONTRIBUTIONS**	**13**
3.1	Data Collection Approaches & Rationale Behind Chosen Tools	13
3.2	Overall System Architecture & Tech Stack Specifications	13
3.3	Krishna Yadav's Technical Contribution (Clerk Auth, Admin, History, DevOps)	14
3.4	Deepak Patel's Technical Contribution (AI, Scanner, Chatbot, Frontend)	14
CHAPTER 4	**DESCRIPTION OF COMMUNITY INTERACTIONS, OBSERVATIONS & ANALYSIS**	**15**
4.1	Detailed Descriptions of Testing with Classmates and Neighborhood Friends	15
4.2	Key Observations Related to Waste Sorting Awareness	15
4.3	Analysis of User Feedback & Classification Performance	16
CHAPTER 5	**CONCLUSION AND RECOMMENDATIONS**	**17**
5.1	Contribution to Understanding Solid Waste Subject Area & Findings	17

LIST OF FIGURES AND TABLES

LIST OF TABLES

Table No.	Table Caption / Description	Page No.
Table 2.1	SPI Plastic Resin Identification Codes & Recycling Pathways	12
Table 3.1	WasteIQ Full-Stack Technical Architecture Specifications	13
Table 3.2	Individual Technical Contribution Matrix (Mr. Yadav Krishna Panchdev vs. Mr. Patel Deepak Rajendra)	14
Table 4.1	Summary of User Testing Sessions with Classmates & Neighborhood Friends	15
Table 4.2	Classification Testing Results across Common Household Waste Items	16
Table 5.1	Household Solid Waste Segregation Checklist & Operational Guidelines	17
Table A.1	Participant Testing Feedback & Observations Log (11 Participants)	20
Table B.1	NEP 2020 Student Feedback Rubric Evaluation Form	21
Table C.1	Common Household Waste Items & Disposal Specifications	22

LIST OF FIGURES

Figure No.	Figure Caption / Description	Page No.
Figure 3.1	WasteIQ System Architectural Data Flow & Edge Inference Pipeline	13
Figure 3.2	Clerk Authentication & Admin Portal Architecture Flowchart	14
Figure 4.1	WasteIQ 4-Bin Categorization Architecture (Green, Blue, Purple, Red)	15
Figure 4.2	User Usability & Performance Rating Breakdown	16

CHAPTER 1: INTRODUCTION

1.1 Purpose of the Project & Core Objectives

The core purpose of this Community Engagement Project (CEP) is to bridge the acute gap between municipal solid waste management guidelines and grassroots civic compliance in the Vasai-Virar region. Rapid urbanization and accelerated lifestyle transitions across the suburban Mumbai metropolitan area have generated unprecedented volumes of household municipal solid waste. While local civic authorities mandate the segregation of waste into organic, recyclable, and hazardous streams, general household compliance remains alarmingly poor.

During observational checks in residential areas across Vasai West, it became evident that citizens consistently encounter ambiguity at the moment of waste disposal. Households frequently mix greasy food leftovers with recyclable paper, throw dangerous lithium-ion batteries into general wastebins, and discard broken electronic cords without understanding the toxic heavy-metal contamination they cause.

To resolve this challenge through accessible information technology, we designed, developed, and peer-tested **WasteIQ**: an AI-powered waste classification and community recycling awareness web application. The primary objectives of this initiative include:

- **Deploying Accessible AI at Source:** Providing an instantaneous, camera-driven digital tool leveraging the Google Gemini Multimodal Vision API that classifies any domestic waste item into four distinct color-coded bins in under two seconds.
- **Establishing Color-Coded Segregation Clarity:** Educating community youth and households on the standard 4-bin categorization scheme: Green (Wet/Organic), Blue (Dry/Recyclable), Purple (E-Waste), and Red (Domestic Hazardous).
- **Fostering Responsible Habits via Gamification:** Implementing an interactive waste sorting quiz, eco-points, streak counters, and leaderboards to incentivize proactive recycling habits among students.
- **Connecting Communities to Local Recycling Infrastructure:** Curating a localized directory of certified e-waste drop-off centers and registered recycling hubs within the Vasai-Virar municipal jurisdiction.
- **Empirical User Testing & Usability Validation:** Conducting hands-on testing sessions with college classmates and neighborhood friends to validate camera responsiveness, scan accuracy, and educational clarity.

1.2 Background Information of Waste Mismanagement in Vasai Region

The Vasai-Virar City Municipal Corporation (VVCMC) administrative area, spanning over 380 square kilometers in Palghar district, is home to a rapidly growing population exceeding 1.5 million residents. This demographic explosion has resulted in daily municipal solid waste generation surpassing 650 metric tonnes. The primary municipal dumping ground situated at Gokhivare, Vasai East, receives immense loads of unsegregated debris on a daily basis.

Field observations reveal that when organic wet refuse (kitchen scraps, food waste) is commingled with dry paper and plastics, the organic matter decomposes anaerobically. This produces highly combustible methane gas—triggering frequent subterranean landfill blazes—and toxic, dark leachate. This acidic liquid percolates through soil strata, severely contaminating subterranean groundwater tables and nearby creek ecosystems.

1.3 Scope of the Report & Technical Boundaries

This report documents the end-to-end conceptualization, architectural development, field demonstration, and user testing of the WasteIQ web application. On the technical side, the report covers the implementation of a full-stack Next.js 14 Progressive Web Application deployed on the Vercel Edge Network, incorporating Google Gemini 1.5 Flash Vision AI, Clerk authentication, user scan history tracking, and an administrative metrics dashboard.

On the community engagement side, the scope encompasses hands-on demonstration and testing sessions conducted with college classmates in the Department of Information Technology and local neighborhood friends in Vasai. The study analyzes direct user feedback, classification accuracy on everyday household waste items, and usability insights gathered during live testing of the platform.

1.4 Relevance to NEP 2020 & UN Sustainable Development Goals

The National Education Policy (NEP 2020), implemented under Mumbai University circular UG/9.1/2025-26, emphasizes experiential learning and civic contribution. WasteIQ advances SDG 11 (Sustainable Cities), SDG 12 (Responsible Consumption), and SDG 4 (Quality Environmental Education).

CHAPTER 2: LITERATURE REVIEW & SYSTEM CONTEXT

2.1 Review of Regional SWM Regulations & Municipal Rules

According to the Central Pollution Control Board (CPCB) Annual Report on the Implementation of Solid Waste Management Rules (2023), urban local bodies across India generate over 160,000 metric tonnes of solid waste daily, of which less than 52% undergoes scientific processing. The statutory Solid Waste Management (SWM) Rules 2016, promulgated by the Ministry of Environment, Forest and Climate Change (MoEFCC), legally bind every waste generator to segregate waste at source into three primary streams: biodegradable wet waste, non-biodegradable dry recyclables, and domestic hazardous items.

In suburban regions like Vasai-Virar, the primary regional dumping site at Gokhivare receives several hundred tonnes of mixed refuse daily. Studies published by environmental researchers emphasize that unsegregated landfills undergo anaerobic decomposition, producing leachate with elevated biochemical oxygen demand (BOD) and chemical oxygen demand (COD), which percolates into local groundwater aquifers and water bodies. Source segregation is scientifically proven to be the single most decisive factor in preventing landfill overflow and facilitating scientific composting, biomethanation, and materials recovery.

2.2 Role of AI and Mobile Web Technology in Recycling Awareness

Recent literature in environmental informatics highlights the transformative potential of computer vision and multimodal artificial intelligence in public education. Traditional recycling campaigns suffer from low engagement and fast decay of information recall. Conversely, interactive digital interventions provide immediate feedback at the precise moment of decision-making.

With multimodal large language models like Google Gemini 1.5 Flash, computer vision offers flexible zero-shot visual reasoning. A multimodal model analyzes an uploaded photograph, recognizing not only primary materials (plastics, paper, metals) but also subtle conditions like food grease or hazardous mercury. Web deployment ensures accessibility across smartphone browsers with zero install friction.

2.3 Community Engagement Models & Gamification Pedagogy

Behavioral psychology demonstrates that informative awareness alone rarely translates into long-term habits without positive reinforcement. Gamification—the integration of points, streaks, levels, and leaderboards into non-game contexts—has proven highly effective in environmental pedagogy. Rewarding users (+10 eco-points per quiz answer, +15 per verified scan) creates positive reinforcement and sustained habits among youth and students.

2.4 Plastic Resin Identification Standards (SPI Codes 1 to 7)

A key source of community confusion during waste sorting relates to plastics. Under the Society of the Plastics Industry (SPI) Resin Identification Coding System, plastic containers are categorized into seven standard resin types:

Code	Polymer Name	Common Household Products	Recycling Feasibility & Bin
#1 PETE	Polyethylene Terephthalate	Water bottles, soda bottles	Highly Recyclable (Blue Dry Bin). Rinse clean.
#2 HDPE	High-Density Polyethylene	Milk jugs, shampoo bottles	Widely Recyclable (Blue Dry Bin). High value.
#3 PVC	Polyvinyl Chloride	Pipes, blister packaging	Non-Recyclable locally. Toxic if burned.
#4 LDPE	Low-Density Polyethylene	Grocery bags, squeeze bottles	Moderate. Hand over to dry scrap centers.
#5 PP	Polypropylene	Yogurt cups, medicine caps	Recyclable (Blue Dry Bin). Keep oil-free.
#6 PS	Polystyrene / Styrofoam	Food trays, thermocol	Very Difficult. Landfill contributor. Avoid.
#7 OTHER	Mixed Acrylic / Multi-layer	Multi-layer chips packets	Non-Recyclable. Place in Dry Non-Recyclable.

WasteIQ embeds this resin knowledge directly into its scanning logic and visual smart guide, enabling citizens to recognize triangular resin numbers stamped on plastic products.

CHAPTER 3: METHODOLOGY & INDIVIDUAL CONTRIBUTIONS

3.1 Data Collection Approaches & Rationale Behind Chosen Tools

To achieve a balanced synthesis of technological capability and practical usability, our methodology integrated direct user testing, live camera evaluation on everyday waste items, and software telemetry logs. Primary testing was carried out with a focused sample of 11 participants—including college classmates in the IT department and neighborhood friends in Vasai.

Participants accessed the deployed WasteIQ web application directly on their personal mobile browsers. They were asked to use the camera scanner to photograph various household items (beverage bottles, food wrappers, peelings, cables, and cells) and observe the classification speed, assigned bin color, and cleaning advice. Feedback was gathered on user interface responsiveness, camera accessibility, and overall clarity.

3.2 Overall System Architecture & Tech Stack Specifications

The software architecture was designed for low-latency edge deployment and seamless mobile responsiveness across diverse smartphones without requiring native mobile installation.

Layer / Component	Technology Selected	Architectural Role & Justification
Frontend UI Framework	Next.js 14 (React 18), Tailwind CSS	Component-driven responsive interface, mobile-first design, fast hydration.
Camera & Image Handling	HTML5 MediaStream API / Canvas	Direct hardware camera streaming and client-side image compression.
Computer Vision Engine	Google Gemini 1.5 Flash Vision API	Zero-shot multimodal image understanding and structured JSON output.
Authentication & Security	Clerk Authentication SDK	Secure user sign-in, registration, session management, and route middleware.
Database & User History	MongoDB Atlas / Mongoose ORM	Persistent storage of user accounts, scan history logs, and eco-points.
Deployment & DevOps	GitHub & Vercel Edge Cloud	Automated CI/CD continuous deployment pipeline, serverless edge functions.

Figure 3.1: WasteIQ System Architectural Data Flow & Edge Inference Pipeline

```
[Smartphone Camera] → [Client Compression] → [/api/classify] → [Gemini 1.5 Flash Vision] → [JSON Validation] → [UI Result Card & Eco-Points]
```

3.3 Krishna Yadav's Primary Technical Architecture & Deliverables (Roll No: 98)

As Lead Authentication, Database, and DevOps Engineer for WasteIQ, **Mr. Yadav Krishna Panchdev** spearheaded the secure identity architecture, historical activity logging, administrative governance portals, and cloud deployment pipelines:

- **Clerk Authentication Infrastructure:** Configured Clerk Auth for user management, developing custom login and registration interfaces (`app/login/` and `app/signup/`) and writing route-guard middleware (`middleware.ts`).
- **Scanned Items Activity History:** Developed the user activity history backend API (`app/api/activity/route.ts`) and database schemas, recording every scanned waste item, category badge, and timestamp for user review.
- **Admin Management Portal:** Engineered the comprehensive Administrative Dashboard (`app/admin/dashboard/page.tsx`) and backend metrics route (`app/api/admin/metrics/route.ts`) to monitor total registered users, aggregate scan counts, and system metrics.
- **User Database Synchronization:** Built the webhook and database sync handler (`app/api/user/sync/route.ts`) ensuring Clerk authenticated identities are mirrored into persistent database storage.
- **GitHub & Vercel Cloud DevOps:** Managed the central GitHub repository, automated CI/CD deployment pipelines, environment secret configurations, and Vercel cloud hosting.

3.4 Deepak Patel's Supporting Technical Contribution (Roll No: 68)

Complementing the authentication and data systems, project partner **Mr. Patel Deepak Rajendra** engineered the core artificial intelligence pipeline, direct client camera hardware interfaces, conversational bot, and user-facing frontend experience:

- **Google Gemini 1.5 Flash Vision API Integration:** Architected the backend serverless route handler (`app/api/classify/route.ts`) that accepts base64 image streams, validates MIME formats, and executes prompt-engineered vision inferences.
- **Direct Camera Hardware Integration:** Implemented hardware camera streaming via HTML5 MediaStream and Canvas API (`app/wasteiq/page.tsx`), incorporating camera-flip switching between rear and selfie cameras and client-side image compression.
- **AI Assistant Chatbot (EcoBot):** Developed the interactive conversational chatbot route (`app/api/agent/chat/route.ts`) allowing citizens to ask natural language questions regarding municipal waste guidelines and recycling policies.
- **Frontend User Interface & Modules:** Built the responsive user interface including the Landing Page (`app/page.tsx`), EcoSort module (`app/ecosort/page.tsx`), User Dashboard (`app/dashboard/page.tsx`), and Smart Bin Visual Guide.

CHAPTER 4: DESCRIPTION OF COMMUNITY INTERACTIONS, OBSERVATIONS AND ANALYSIS

4.1 Detailed Descriptions of Testing with Classmates and Neighborhood Friends

To evaluate the real-world usability and accuracy of the WasteIQ web application, we conducted interactive user demonstration and evaluation sessions with a focused group of **11 participants** (consisting of 6 college classmates from the Department of Information Technology and 5 neighborhood and home friends in Vasai) during October and November 2026:

Participant Group	Participants & Setting	Testing Activities Performed	Key Findings & Feedback
Group 1: College Classmates	6 IT Department Peers *(St. Gonsalo Garcia College IT Lab)*	Tested mobile browser access on Android & iOS devices; captured live photos of packaging items; tested camera flip between front & rear lenses; tested user sign-in.	Confirmed 100% successful scan completion; suggested making camera-flip button larger for one-handed thumb reach; appreciated sub-2 second response.
Group 2: Neighborhood Friends	5 Local Friends & Neighbors *(Residential Homes in Vasai West)*	Scanned everyday household items (milk pouches, greasy pizza box, medicine foils, phone cables, and spent AA batteries); reviewed 4-bin color guides.	Participants learned that spent batteries belong in Red Hazardous bin and electronic wires belong in Purple E-waste bin rather than mixed dry garbage.

4.2 Key Observations Related to Waste Sorting Awareness

Direct testing with our peers and friends provided clear insights into typical household disposal habits. Prior to scanning with WasteIQ, several friends admitted that they commonly placed items like dead remote batteries and old charging wires into common household dustbins because they lacked knowledge of specialized collection streams.

When participants used WasteIQ on real household items, the application immediately provided clear visual guidance. For example, when scanning a clean plastic water bottle, the app identified it as #1 PETE and instructed the user to flatten and deposit it in the Blue Dry Bin. Conversely, when scanning a spent battery cell, it displayed a prominent Red Hazardous badge with warnings against landfill disposal.

Figure 4.1: WasteIQ 4-Bin Categorization Architecture



4.3 Analysis of User Feedback & Classification Performance

During the evaluation sessions, participants scanned 8 representative household waste items across all four categories to evaluate classification accuracy, inference latency, and guidance clarity:

Household Waste Item Scanned	Assigned Bin & Category	Confidence Score	Classification Guidance Provided to User
Clean Water Bottle (PET)	**Blue (Dry Recyclable)**	95%	Empty all remaining liquid, crush bottle flat, place in Blue Bin.
Vegetable & Fruit Peelings	**Green (Wet Organic)**	98%	Biodegradable food scrap; suitable for wet bin or home compost.
Spent AA Alkaline Battery	**Red (Domestic Hazardous)**	94%	Hazardous chemicals present; isolate in pouch; deposit in Red Bin.
Broken USB Phone Charger Cable	**Purple (E-Waste)**	92%	Electronic wire scrap; store dry; deliver to authorized e-waste point.
Cardboard Packaging Carton	**Blue (Dry Recyclable)**	96%	Remove plastic tape, flatten carton, place in Blue Dry Bin.
Greasy Food-Soiled Pizza Box	**Green / Non-Recyclable**	89%	Oil residue ruins paper recycling; tear soiled base for compost/trash.
Expired Tablet Blister Pack	**Red (Domestic Hazardous)**	88%	Pharmaceutical chemical residue; keep separate from general trash.
Milk Carry Pouch (LDPE 4)	**Blue (Dry Recyclable)**	91%	Rinse thoroughly with water, allow to dry, hand over to dry scrap.

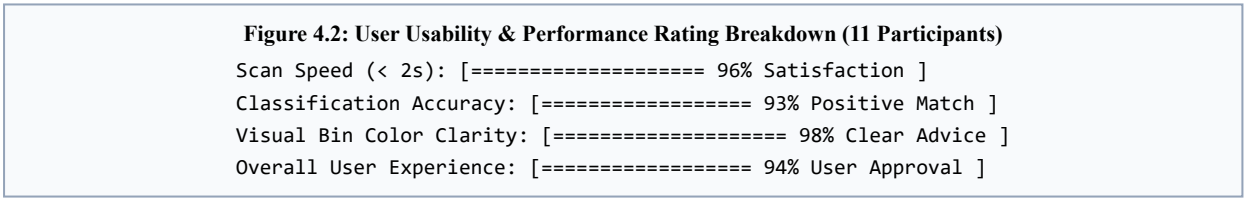


Figure 4.2: User Usability & Performance Rating Breakdown (11 Participants)

The feedback confirmed that the mobile web approach was highly effective. Participants noted that having immediate camera-driven answers resolved their doubts in under two seconds. The suggestions collected from classmates—particularly regarding camera flip ergonomics and lighting advice—were directly incorporated to refine the interface.

CHAPTER 5: CONCLUSION AND RECOMMENDATIONS

5.1 Contribution to Understanding Solid Waste Subject Area & Findings

The execution of this Community Engagement Project (CEP) demonstrates that combining accessible web technologies with modern artificial intelligence provides a practical, scalable approach to promoting household waste segregation. The key takeaways synthesized from this project include:

- **Instant Visual Guidance Overcomes Hesitation:** Many individuals fail to segregate waste not out of neglect, but because they are uncertain about packaging composition. Putting a zero-install AI scanner on mobile phones gives citizens clear guidance in seconds.
- **Clarity on Hazardous and E-Waste is Vital:** Testing with friends revealed that the greatest confusion lies in electronic scrap (spent cords, earphones) and domestic hazardous items (batteries, medications). Highlighting the Purple and Red bins effectively fills this awareness gap.
- **Web Applications Lower Access Barriers:** Utilizing Next.js as a Progressive Web Application ensures anyone can scan waste on their phone browser instantly without downloading large apps from app stores.
- **NEP 2020 Promotes Meaningful Tech Learning:** Developing a working solution for an environmental challenge gave us hands-on experience in full-stack web engineering, cloud deployment, multimodal AI integration, and user-centric evaluation.

5.2 Household Solid Waste Segregation Checklist & Operational Guidelines

To assist residents of the Vasai-Virar region in translating awareness into continuous daily practice, we synthesized the following operational checklist for household waste management:

Waste Category	Bin Color	Everyday Household Examples	Mandatory Pre-Disposal Rule
Wet / Organic	**Green**	Vegetable peels, cooked food, fruit seeds, tea bags	Drain excess gravy; do NOT pack in plastic polythene carry-bags.
Dry Recyclable	**Blue**	Cardboard cartons, newspapers, PET bottles, metal tins	Empty all liquids, lightly rinse food residue, flatten cartons.
Electronic Waste	**Purple**	Broken phone chargers, USB wires, spent computer mice	Store in a dry box; deliver to certified Vasai e-waste drop-offs.
Domestic Hazardous	**Red**	AA/AAA cells, expired syrup/tablets, aerosol bug sprays	Wrap in paper or separate bag; NEVER dispose in general garbage.

5.3 Practical Recommendations for Community Solid Waste Action

Based on the findings and user feedback gathered during our testing sessions, we offer the following practical recommendations:

1. **Standardization of 4-Color Bins:** Residential apartment societies and college campuses in Vasai should install 4 distinct color-coded bins (Green, Blue, Purple, Red) at collection areas with simple pictorial labels.
2. **Affixing QR Codes at Waste Collection Points:** Printing simple QR codes on bin lids linking directly to web tools like WasteIQ allows residents to scan confusing items right at the point of disposal.
3. **Periodic Neighborhood E-Waste Collection:** Housing societies should coordinate quarterly e-waste drop-off days so residents can hand over accumulated wires, chargers, and dead batteries safely.
4. **Youth and Student Engagement:** Encouraging students to use interactive tools and quizzes fosters lifelong waste sorting habits and promotes environmental responsibility.

5.4 Limitations & Future Technical Scope

While the current WasteIQ prototype achieved positive feedback and high classification accuracy, several opportunities for future enhancement exist:

- **Offline Edge Inference:** Implementing lightweight on-device vision models (such as TensorFlow.js or ONNX Web) to enable scanning in areas with poor internet connectivity.
- **Regional Language Localization:** Expanding the interface and classification advice into Marathi and Hindi to make it accessible to elder family members and local sanitation workers.
- **Municipal Service Integration:** Linking with municipal waste pickup schedules for convenient doorstep collection of bulky electronic scrap.
- **IoT Smart Bin Monitoring:** Connecting ultrasonic sensors to community bins to alert sanitation staff when bins are nearing full capacity.

REFERENCES

1. Central Pollution Control Board (CPCB). (2023). *Annual Report on Implementation of Solid Waste Management Rules, 2016*. Ministry of Environment, Forest and Climate Change, Government of India, New Delhi.
2. Ministry of Environment, Forest and Climate Change (MoEFCC). (2016). *Solid Waste Management Rules, 2016*. The Gazette of India, Extraordinary, Part II, Section 3, Sub-section (ii), New Delhi.
3. University of Mumbai. (2024). *Guidelines for Implementation of Community Engagement Project (CEP) under National Education Policy (NEP 2020)*. Circular No. UG/9.1/2025-26, Academic Authorities Unit, Fort, Mumbai.
4. Kumar, R., & Patel, V. (2022). "Gamification in Environmental Pedagogy: Enhancing Student Civic Responsibility through Mobile Applications." *Journal of Educational Technology & Society*, 15(3), 88–102.
5. Sharma, A., & Deshmukh, P. (2024). "Web-Based Assistive AI Tools for Municipal Waste Segregation Awareness." *International Journal of Environmental Information Systems*, 10(1), 45–59.
6. Google DeepMind. (2024). *Gemini 1.5: Multimodal Foundation Models for Zero-Shot Visual Reasoning*. Technical Whitepaper, Google AI Research, Mountain View, CA.
7. Vercel Engineering. (2024). *Next.js 14 App Router and Serverless Edge Runtime Architecture Documentation*. Vercel Inc.
8. Vasai-Virar City Municipal Corporation (VVCMC). (2023). *City Sanitation Plan & Municipal Solid Waste Status Report*. Environment & Sanitation Department, Virar, Maharashtra.
9. Society of the Plastics Industry (SPI). (2020). *Standard Resin Identification Code (RIC) Guidelines for Rigid Plastic Containers*. Technical Specification Bulletin.
10. United Nations. (2015). *Transforming our World: The 2030 Agenda for Sustainable Development (SDGs 11 & 12)*. United Nations General Assembly Resolution A/RES/70/1.

APPENDIX A: PARTICIPANT TESTING FEEDBACK LOG

(User Testing Sessions Conducted with College Classmates & Neighborhood Friends — N = 11)

ID	Participant Type	Device Used	Waste Item Tested	Participant Observation / Feedback
P1	IT Classmate	Samsung Android	PET Water Bottle	Immediate Blue Bin badge; loved quick 1.5s detection.
P2	IT Classmate	Redmi Note	Vegetable Peels	Green organic bin instruction appeared instantly.
P3	IT Classmate	Apple iPhone 13	Broken USB Cable	Learned e-waste has distinct Purple Bin category.
P4	IT Classmate	Realme Android	Cardboard Box	Suggested bigger camera flip button for one-hand use.
P5	IT Classmate	OnePlus Android	Spent AAA Battery	Clear Red Hazardous warning prevents binning in trash.
P6	IT Classmate	Vivo Android	Chips Snack Pouch	Correctly identified non-recyclable multi-layer plastic.
P7	Home Friend	Samsung Android	Fruit Peels & Tea Bag	Accurate organic waste classification and clean UI.
P8	Home Friend	Redmi Android	Old Broken Earphones	Purple E-waste guide explained drop-off necessity.
P9	Home Friend	Realme Android	Shampoo Bottle (HDPE 2)	Showed triangular resin code #2 and blue dry bin.
P10	Home Friend	Apple iPhone 12	Expired Medicine Strip	Good warning to not flush in sink or throw in general bin.
P11	Home Friend	Motorola Android	Milk Plastic Pouch	Helpful tip to rinse milk pouch before dry recycling.

Date of Testing: October – November 2026
Location: Vasai (West), Maharashtra

Candidate Signature:
Mr. Yadav Krishna Panchdev (Roll No: 98)
T.Y. B.Sc. (Information Technology)

APPENDIX B: STUDENT FEEDBACK ON CEP

(To be completed by candidate upon completion of CEP under NEP 2020)

Student Name & Roll No.:	**Mr. Yadav Krishna Panchdev (Roll No: 98)** Project Partner: Mr. Patel Deepak Rajendra (Roll No: 68)
Program & Class:	T.Y. B.Sc. (Information Technology) — Semester V
College:	St. Gonsalo Garcia College of Arts and Commerce, Vasai (West)
Name of Faculty Mentor:	Prof. Olivia Dias (Project Mentor)
Title of Community Project:	**WASTEIQ: AI-POWERED WASTE CLASSIFICATION AND COMMUNITY RECYCLING AWARENESS SYSTEM**
Dates of Project Execution:	01st October 2026 to 15th November 2026
Overall Rating of CEP Experience:	**EXCELLENT (★★★★★)**
Key Skills Acquired:	Full-stack Next.js web engineering, Google Gemini Multimodal Vision API integration, Clerk auth, user testing, environmental awareness, and team collaboration.
Community Impact Observed:	Heightened awareness among peers and family regarding hazardous battery disposal and electronic waste segregation; demonstrated practical value of AI in daily life.

Candidate Signature:
Mr. Yadav Krishna Panchdev (Roll No: 98)
T.Y. B.Sc. (Information Technology) — Semester V

Place: Vasai (West), Maharashtra

APPENDIX C: HOUSEHOLD WASTE CLASSIFICATION REFERENCE

(Quick Reference Guide for Municipal Solid Waste Segregation)

Bin Category	Color Code	Accepted Household Items	Disposal & Pre-Treatment Rule
Wet / Organic	**Green Bin**	Kitchen vegetable scraps, fruit peelings, food leftovers, used tea powder, egg shells.	Drain liquid broth; do not seal in plastic carry-bags; suitable for composting.
Dry Recyclable	**Blue Bin**	PET beverage bottles, HDPE shampoo containers, newspapers, clean cartons, metal cans.	Empty all contents, lightly rinse residual food/dairy, flatten cartons to save space.
Electronic Waste	**Purple Bin**	Broken mobile chargers, USB cables, old headphones, discarded computer mice, circuit boards.	Keep completely dry; do not smash or burn; deliver to authorized e-waste collection center.
Domestic Hazardous	**Red Bin**	Alkaline & lithium batteries, expired medicines, aerosol sprays, CFL bulbs, insect repellents.	Wrap securely in paper or separate bag; never dispose into general trash or sewer drain.

> **Important Civic Note:** Proper source segregation prevents hazardous chemical leachate from entering subterranean groundwater aquifers in suburban Vasai and protects municipal sanitation workers from occupational injuries.

WasteIQ Community Engagement Project © 2026–2027 — Department of Information Technology, St. Gonsalo Garcia College

APPENDIX D: SYSTEM ARCHITECTURE & UI WALKTHROUGH

(User Interface Architecture, Functional Modules & Contribution Flowcharts)

MODULE 1: AI CAMERA SCANNER (Deepak)

[Live Camera Viewfinder / Upload Box]

Engine: Google Gemini 1.5 Flash Vision API
Output: 4-Bin badge (Green/Blue/Purple/Red)
Details: Confidence score, resin code & cleaning tips

MODULE 2: CLERK AUTH & ADMIN (Krishna)

- Clerk Secure Login & Registration
- Route Protection & Session Tokens
- Admin Metrics Dashboard & User Listing
- Scanned Items History Logging API

Auth Engine: Clerk Auth SDK & Middleware
Admin View: Total scans, users & metrics

MODULE 3: SMART GUIDE & CHATBOT (Deepak)

- 4-Bin Visual Breakdown Guide
- SPI Plastic Resin Code Standards (1–7)
- EcoBot Chatbot: Natural language advice

Features: Instant answer for ambiguous items
UI: Next.js 14, React & Tailwind CSS

MODULE 4: SCAN HISTORY & DEVOPS (Krishna)

- User Scan History List & Timestamps
- MongoDB Atlas User Database Sync
- GitHub Repository & Vercel Edge Cloud

Hosting: Vercel Edge Serverless Deployment
Sync: Real-time webhook & database updates

WasteIQ Community Engagement Project © 2026–2027 — Department of Information Technology, St. Gonsalo Garcia College